

Project "Continuous Flow"

Candidate Case Study - Packaging Line Reliability

Assessment objective

This case is designed to assess **structured problem** solving, learning agility, prioritization, and clarity of communication in an unfamiliar manufacturing context.

You will be evaluated basis your originality, research and structured problem solving. There is no right or wrong solutions, we only expect to see your approach and problem solving acumen.

You can expect to solve this case in 45-60 minutes of dedicated time.

Instructions for candidates

- Please take time to read the case, research unfamiliar ideas if needed, and prepare your response.
- Submit: (i) a 3 slider (PPT format) solution with an additional slide containing – Name, Contact details (Phone and Email), Campus Name and Program name (E.g. B. Tech Mechanical)
- You may make reasonable assumptions, but please state them clearly.
- Submission Guideline- You shall submit the file with name format- Campus_YourFullName

Business context

As a summer intern at a large chocolate manufacturing plant, you quickly learn the critical role of the packaging lines in maintaining overall operations. The facility operates three such lines, but Line 2 stands out as the most vital: it handles the plant's highest-volume product—premium chocolate bars—and serves as the primary bottleneck for the entire production process.

Line 2 packages these bars at an impressive rate of 1,000 per minute. However, its single point of vulnerability lies in one unstable machine, which, if it stops for too long, forces upstream production lines to slow or halt entirely. A single disruption here can cascade through the plant, threatening daily output targets and profitability.

How the line works

Chocolate bars arrive -> Bar is wrapped -> Wrapper is sealed -> Wrapped bar moves into cartoning -> Finished Goods are Dispatched

Key components

Component	What it does
Photo-eye sensors	Detect marks on the wrapper film so the machine knows where to cut and seal.
Sealing jaws	Heated jaws that seal the wrapper around the chocolate bar.
Servo motors	Drive robotic movement that transfers wrapped bars into the cartoning section.

The problem

A few months ago, Line 2 was performing at a very high level. Its Overall Equipment Effectiveness (OEE) was around 98%. Recently, erratic stoppages and breakdowns have caused OEE to fall to 72%. The maintenance team is spending most of its time reacting to failures.

To reduce downtime, the maintenance team introduced a simple rule: replace all critical sub-components every 150 operating hours. However, this has not improved reliability. Spare part costs have increased, and many replaced parts still appear healthy.

Failure patterns observed

Component	Observed pattern
Sensors	Some fail within a week; some last for many months. Failures appear unpredictable. Possible causes include dust and power fluctuation.
Sealing jaws	Most failures happen around 300 operating hours. Failures appear linked to wear and heat over time.
Servo motors	Many newly installed motors fail within the first 40 operating hours. Motors that survive beyond this period usually run well for months.

Additional information

- You believe that the current downtime data is not being captured in enough detail.
- Some stops are minor- stops (short stoppages usually cleared by operators), while others are major breakdowns (longer stoppages requiring maintenance intervention).
- The quality team also believes the sealing jaw problem is linked to temperature control.

Sealing jaw temperature data	Value
Target temperature (T)	140°C
Lower Spec Limit (LSL)	135°C
Upper Spec Limit (USL)	145°C
Current mean (μ)	143°C
Standard deviation (σ)	1.0°C

Questions

Question 1: Diagnostic structuring and data architecture (20 Marks)

- Design a structured data collection approach to identify the real reasons for the OEE drop from 98% to 72% while addressing the below questions-
 - What data would you collect?
 - How would you classify stoppages?
 - How would you separate minor - stops from major breakdowns?
 - How would this help isolate the main causes of loss?

Hint: Try to think about the different ways to collect and analyze the data that would help you Define, Measure, Analyze, Improve and Control the OEE.

Question 2: Reliability and maintenance strategy (15 Marks)

The plant is currently replacing all critical components every 150 operating hours.

- Using the concept of the Bathtub Curve, map the three components (Sensors, Sealing Jaws, Servo Motors) to their respective failure phases (Infant Mortality, Random, Wear-out).
- Why is the blanket 150 - hour replacement strategy flawed? What maintenance approach would you recommend for each component? Why?

Hint: Try to find out different maintenance approaches depending on failure types and suggest the best suited approach for each equipment type and give you rationale.

Question 3: Process capability and reliability (15 Marks)

The sealing jaws must maintain the correct temperature to avoid bad seals, film burning, jams, and premature wear.

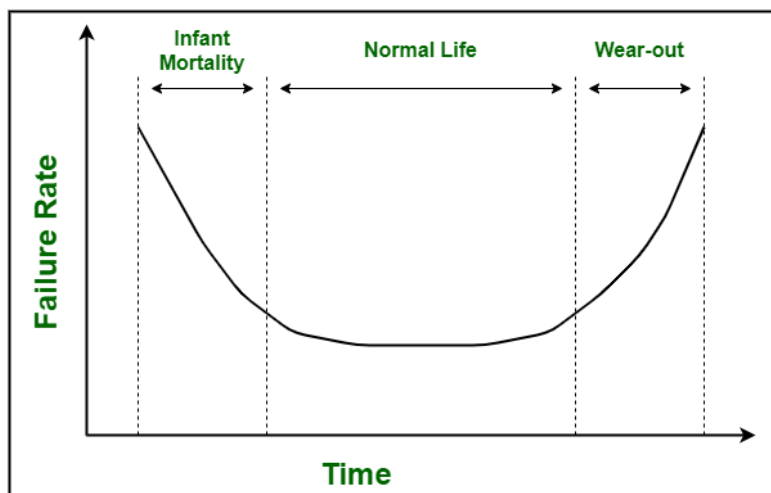
Part A: Calculate Cp and Cpk. Then explain whether the process is capable, whether it is centered, and why jams and wear may still happen.

Part B: Calculate Cpm. Then explain why Cpm is useful here and why being away from target can still damage the machine even if the process remains within specification.

Hint: Think carefully about the difference between meeting specification and operating around the ideal target.

Glossary

Bathtub Curve:



Bottleneck: The most capacity-constrained step in the process. If it stops, the whole system is affected.

OEE: Overall Equipment Effectiveness. A measure of machine performance using Availability, Performance, and Quality (Availability*Performance*Quality)

Availability = Actual run time/ Planned Run Time (For Eg. If 1 shift =8 hrs and machine was in breakdown for 1 hour- availability will be $7/8 = 87.5\%$)

Performance = Actual Output/Theoretical Output (Eg. If line capacity is 1000kg/hr, actual output is 800kg/hr then Performance =80%)

Quality= Good Product Output/Actual Product Output (E.g If line produce 1000 bars and 50 bars are sent for rework or scrapped, then Quality= $950/1000 = 95\%$)

Minor-stop: A short stoppage, usually cleared by operators under 10 minutes

Major breakdown: A longer stoppage requiring maintenance support, this takes more than 10 minutes to repair.

Root Cause Analysis: A structured way to find the underlying cause of a recurring problem.

Cp: Shows whether the process spread fits inside specification limits.

Cpk: Shows whether the process is both capable and centered relative to the nearest limit.

Cpm: Shows whether the process is capable while also staying close to the target.

Formula sheet

$$Cp = (USL - LSL) / (6\sigma)$$

$$Cpk = \min[(USL - \mu)/(3\sigma), (\mu - LSL)/(3\sigma)]$$

$$Cpm = (USL - LSL) / [6\sqrt{\sigma^2 + (\mu - T)^2}]$$

End of case study